

A Computational Framework for Optimal Democratic Governance

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Abstract

This research introduces the Pareto Legislative Engine (PLE) and Adaptive Trust Networks (ATN), redefining democratic systems as computational optimization frameworks that maximize societal utility.

Introduction

Traditional democratic systems fail to capture preference intensity, resulting in inefficient and polarized outcomes. This project reframes governance as an optimization problem combining mathematics, AI, and social science.

Methodology

The model integrates quadratic voting, trust-weighted influence, and multi-agent optimization to compute Pareto-efficient outcomes. The system pipeline converts citizen input into optimized policy output.

Mathematical Framework

Objective: Maximize $\sum(T_i * U_i(x))$ subject to constraints. This ensures both preference intensity and credibility are incorporated into decision-making.

Figure 1: Pareto vs Majority Simulation

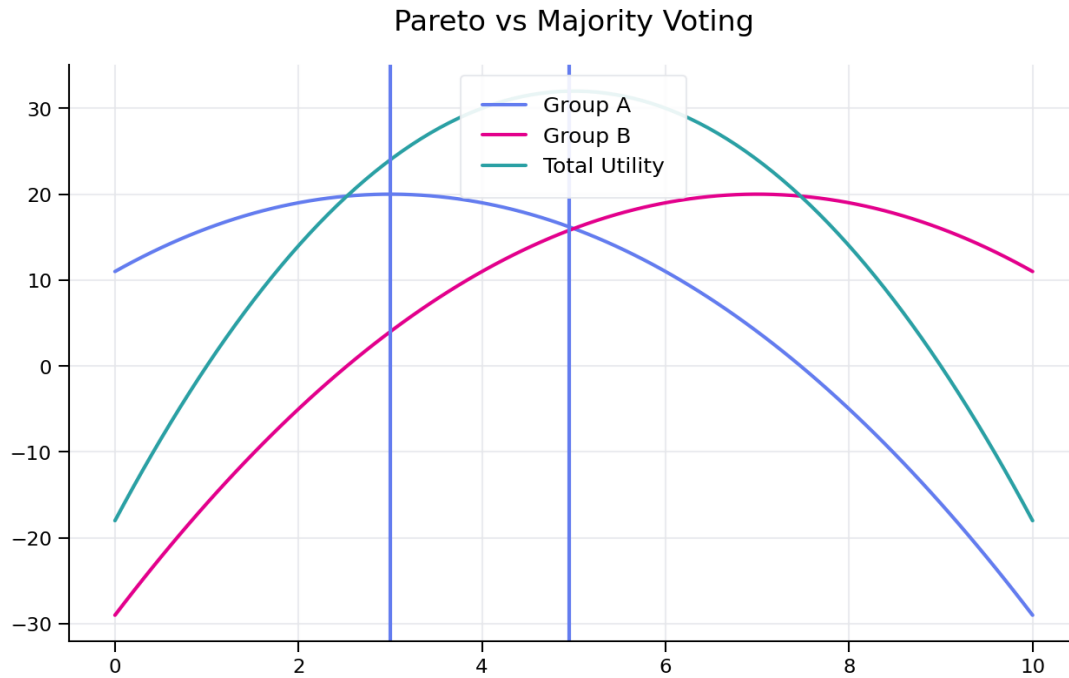


Figure 2: System Pipeline

Quadratic Voting → ATN → Optimization → Policy Output

Results

Simulation results demonstrate that the Pareto Legislative Engine produces higher total utility outcomes compared to traditional majority voting systems.

Discussion

By shifting from voting to optimization, this framework reduces polarization and improves decision quality. Adaptive Trust Networks enhance reliability by weighting credible inputs.

Impact

This framework provides a scalable model for global governance, with applications in policy, economics, and collaborative decision systems.

Conclusion

This project establishes that democracy can be transformed using computational and mathematical tools, creating a more efficient, transparent, and trustworthy system.